

INFECTIOUS GRANULOMA

DR EMAN GOMAA

CONSULTANT DERMATOLOGIST

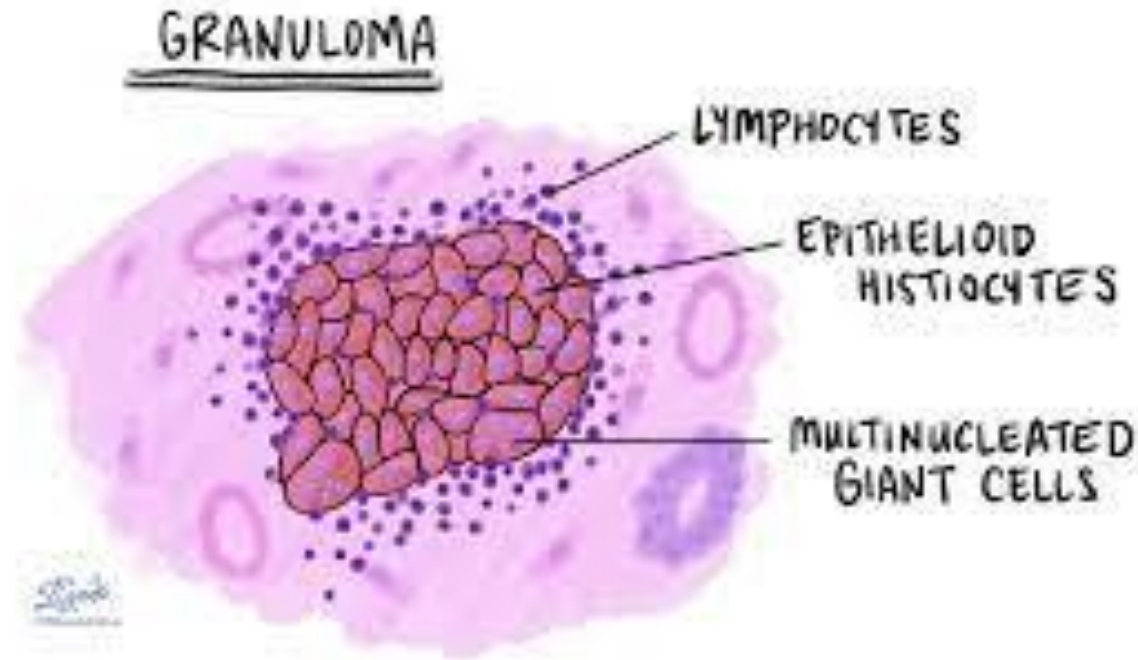
Granuloma

granuloma:- chronic inflammatory condition of skin in which histiocytes [tissue macrophages] are predominant cells.

Histiocytes →

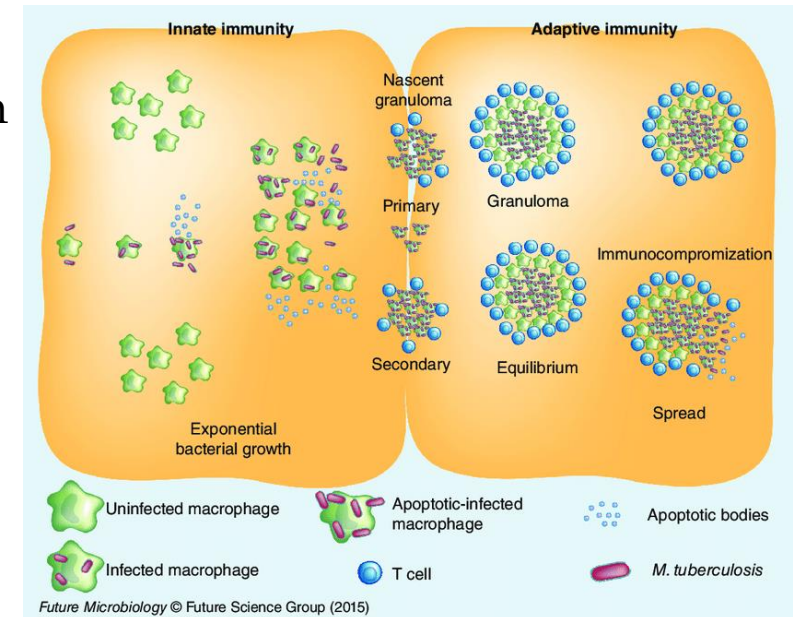
professional phagocyte (CD 68) →

APC Langerhans cells



Steps of granuloma

- Type IV hypersensitivity
- Immune response to persistent antigen → indigestible.
- Engulfed by histiocytes
- Immobilization → accumulation → epithelioid cyst →
- multinucleated giant cells →
- surrounded by lymphocytes, fibroblasts with central caseation



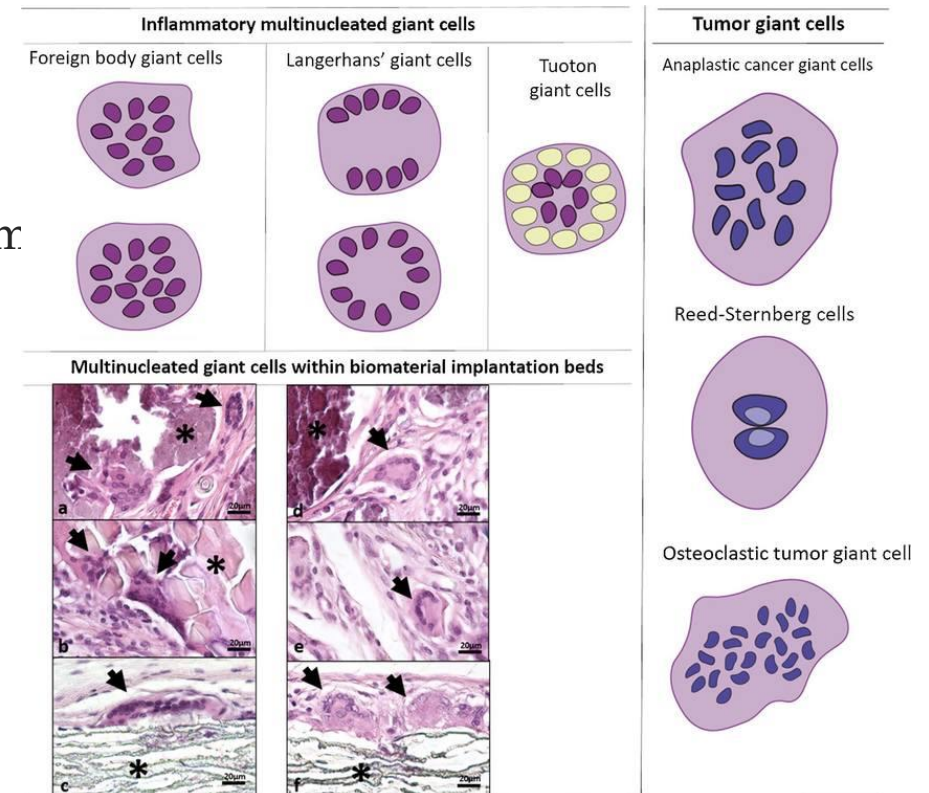
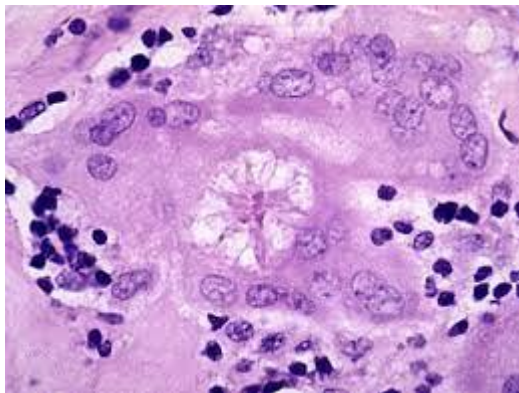
Types of giant cells

Langhans:- horseshoe, peripheral & tubercloid

Touton:- circle, center, foamy & xanthoma.

Foreign body:- paphazurd

Sarcoidal:- langhans type asteroid (stellate) & Schaum



Types of granuloma:-

1] **Tubercloid** :- langhans / dense rim of lymphocyte / caseation → T.B/ tubercloid leprosy & LMDF

2] **Sarcoidal** [necked] granuloma :- asteroid & shumann body.
Sparse lymphocyte → sarcoidosis / sarcoidal reaction

3] **palisading granuloma**:- palisade like around necrobiosis/ collagen
GA, NBL, RN, AEGCG/ necrobiotic xanthogranuloma

4] **foreign body granuloma**

5] **suppurative**:- neutrophils & infectious agent

MYCOBACTERIA

- Mycobacteria:-

Similar fungal molds

Bacilli Slowly growing Lipid cell wall

G +ve Acid fast / alcohol fast

Obligate intracellular pathogens

Mycolic acid → most important lipid

190 species → 25 pathogenic to human

M. leprae, *M. tuberculosis*

M. Marinum, *M. ulceration*

M. Kansasii & *M fortuitum*

MYCOBACTERIA THAT CAUSE CUTANEOUS DISEASE		
Group and pigment	Rate of growth	Pathogens
Slow growers		
Photochromogens*	2–3 weeks	<i>M. kansasii</i> , <i>M. marinum</i> , <i>M. simiae</i> <i>M. scrofulaceum</i> , <i>M. szulgai</i> , <i>M. gordonae</i> , <i>M. xenopi</i> <i>M. tuberculosis</i> , <i>M. avium</i> , <i>M. intracellulare</i> , <i>M. ulcerans</i> , <i>M. haemophilum</i> , <i>M. malmoense</i> , <i>M. terrae</i> , <i>M. genavense</i> , <i>M. bovis</i> §, <i>M. nonchromogenicum</i>
Scotochromogens†	2–3 weeks	
Nonchromogens‡	2–3 weeks	
Rapid growers	3–5 days	<i>M. fortuitum</i> , <i>M. chelonae</i> , <i>M. smegmatis</i> , <i>M. abscessus</i> **, <i>M. immunogenum</i> , <i>M. goodii</i> , <i>M. wolinskyi</i> , <i>M. cosmeticum</i> , <i>M. mucogenicum</i> , <i>M. mageritense</i>
Non-cultured (to date)		<i>M. leprae</i>

M. tuberculosis

- Pathogen:- tuberculosis
- Staining:- **difficult / resistant decolorization**
- Acid fast:- 25% sulphuric acid
- Alcohol fast:- yes
- Special stain:-Ziehl-Neelson
- Culture:- Lowenstein Jensen

M.leprae

- Leprosy
- More easily / less resistant.
- 5% sulphuric
- No
- Lite stain
- None

Histiocyte receptor

- Carbohydrate → lectin → foreign material.
- Complement → CR, CRs
- FC fragment IGg

Leprosy Hansen's disease

- Chronic slowly progressive infectious disease caused by *M. leprae* → predilection to → Peripheral nerve, skin, mucosa and internal organs

- *M. leprae*

Acid fast

Obligate intracellular

Macrophages / Schwann

Cold area 35%

Peripheral N

Viable 10 days in environment V. hard

Slowly growing Fite stain no culture → mouse foot pad & nine banded armadillo

- Epidemiology:-

all age → 10-15 / 30-60

Equal sex / 86% in 6 countries

India 60%

Aetiopathogenesis

- Source:- cutaneous leprosy patient
- Mode:- discharge/droplet & eroded skin → direct inoculation & nasal secretions
- I.P:- month 30 – yrs [4-10 yrs]
- Factors determine infectivity:-
 1. Contagious patient
 2. Susceptible person
 3. Close contact

HLA DR2-DR3 → tuberculous

HLA DP 1 → lepromatous

- **Pathogenesis/ outcome of infection:-**

Inhaled → multiplies in nose → circulation → Schwann cell/macrophages → cytokin IL 1- IL 12 –TNF alpha → increased number & activity of macrophages

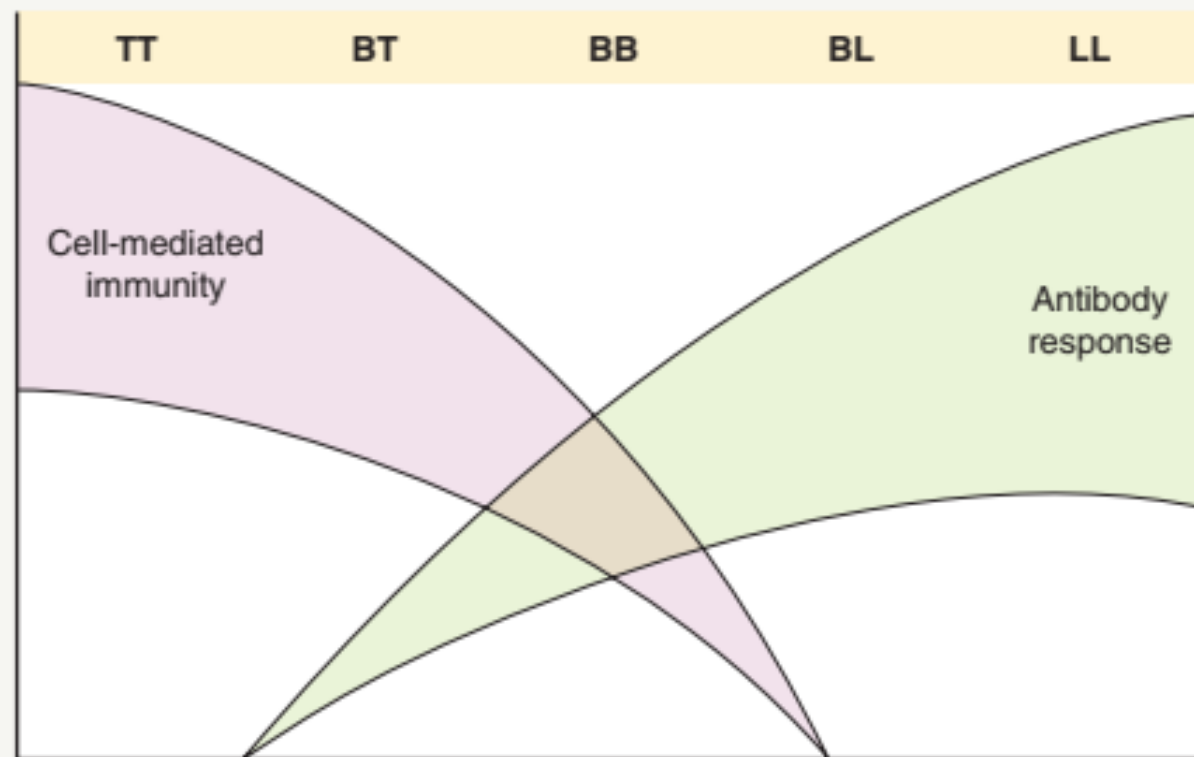
Classification :-

Ridly, Jobling classification [5] TT, BT, BB, BL & LL

WHO classification :PAUCIBACILLARY , MULTIBACILLARY

Rabella class [3]

THE CLINICAL-IMMUNOLOGIC SPECTRUM OF LEPROSY



Tissue expression of cytokines

IL-2, IFN- γ

IL-4, IL-10

M. leprae in tissues

Leprosy reactions
Type 1 reactions

upgrading

Type 2 reactions

CLASSIFICATION OF LEPROSY						
Clinical findings	LL	BL	BB	BT	TT	I
Type of lesions	Macules, papules, nodules, diffuse infiltration	Macules, papules, plaques, infiltration	Plaques and dome-shaped, punched-out lesions	Infiltrated plaques	Infiltrated plaques, often hypopigmented	Macules, often hypopigmented
Number of lesions	Numerous	Many	Many	Single, usually with satellite lesions, to more than 5	One or few (up to 5)	One or few
Distribution	Symmetric	Tendency to symmetry	Evident asymmetry	Asymmetric	Localized, asymmetric	Variable
Definition	Vague, difficult to distinguish normal versus affected skin	Less well-defined borders	Less well-defined borders	Well-defined, sharp borders	Well-defined, sharp borders	Not always defined
Sensation	Not affected	Diminished	Diminished	Absent	Absent	Impaired
Bacilli in skin lesions	Many (globi)	Many	Many	Few (1+), if any, detected	None detected	Usually none detected

C/pic

- **Tubercloid leprosy**

- Strong immunity
- Th1 → IL 2- IFN – TNFb → maintain inflamm. → well formed tubercloid granuloma
- Paucibacillary → nerve – skin

No mucosal or organ affection

- **Lepromatous leprosy**

- Weak immunity
- Th2 → iL 4,5,10 –IL b → suppress Macrophages → diffuse tubercloid granuloma
- Multibacillary → skin-mucosal-late organ and nerve late affection

Tubercloid

- Skin lesion:-
 - Solitary or few asymmetrical
 - Well demarcated macule or plaque
 - Elevated sloping border
 - Erythematous or hypopigmented
 - [never depigmented]
 - Dry, hairless & insensitive

Early nerve affection



lepromatous

- Multiple, bilateral
- Symmetrical poorly defined
- Erythematous macule, papule, plaque and nodule
- Face/ buttocks / UL & LL
- Not affected:- high temperature area
- Hairy scalp, axilla, groin & perineum

Late nerve affection





Fig. 75.7 Tuberculoid leprosy. **A** Plaque with slightly elevated, erythematous border and central hypopigmentation. **B** Large annular lesion on the cheek with a red-brown color and rim of hypopigmentation.



eFig. 75.1 Lepromatous leprosy. Numerous nodules on the forearms. Courtesy Louis Fragola, MD.



Fig. 75.4 Lepromatous leprosy. **A** Numerous erythematous papules and nodules on the forearms and hands. **B** Infiltrated nodules coalescing on the forehead with leonine facies and madarosis. Note the ocular involvement.

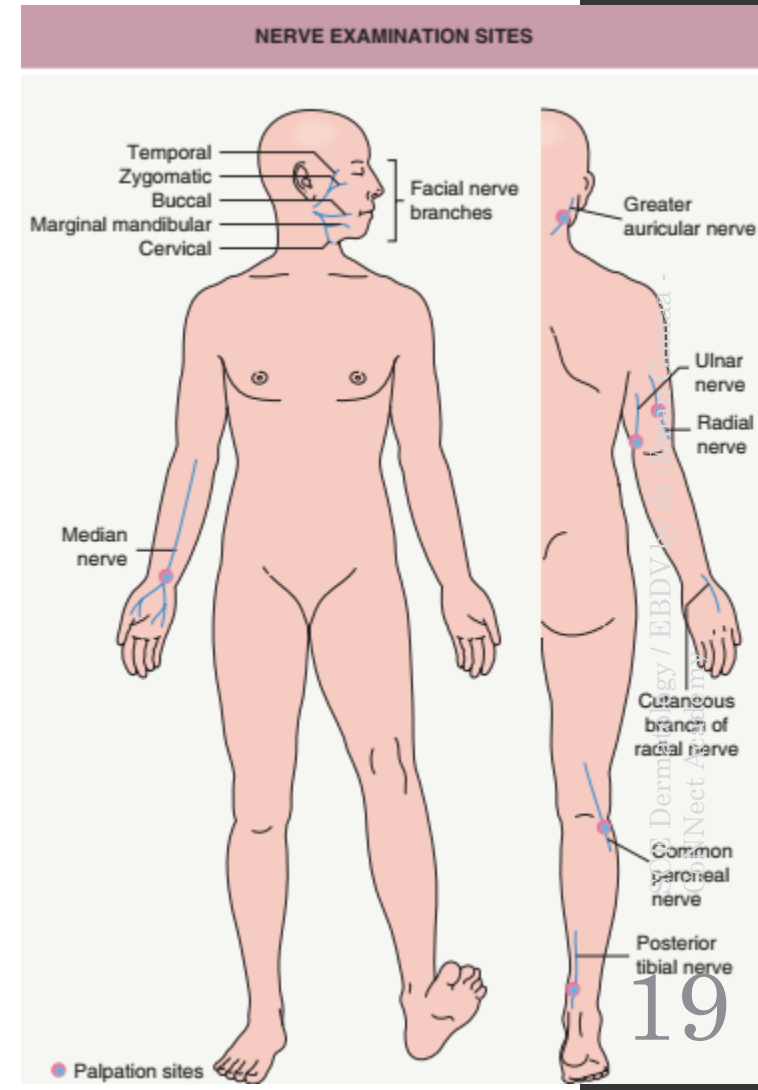


Fig. 2.2.2.2. Leprosy. A, B Diffuse infiltration with leonine facies and madarosis. A, Courtesy, Louis A Fragola, Jr, MD.



NERVE AFECTION

- Peripheral N + superficial location
- Post-tibial, ulnar, median, lateral popliteal and facial nerves.
 - Sensory:- anathesia, hypothesia [decreased pain, temperature and touch sensation]
 - Motor:- weakness, paralysis, ms atrophy & neuropathic changes
 - Autonomic :- hairless, dryness, hypopigmented & vasomotor alternation



• In tubercloid leprosy

- No mucosal
- No internal organ affection because of strong immunity



Lepromatous leprosy

- Hyperaemic, ulcerated nasal mucosa → epistaxis
- Oral papule → lips, palate, uvula, tongue and gum
- Nerve affection → numbness, anaesthesia on dorsum of hand & feet while sensation in palm, sole retained

Sequelae / internal organ affection:-

Leonine facies

Madarosis

Eye :- lagophthalmos, corneal anaesthesia

Saddle nose, upper incisor teeth loosen

Hoarsness of voice / gynecomastia

Hand & feet →

papal hand [ulnar] 4th & 5th contracture

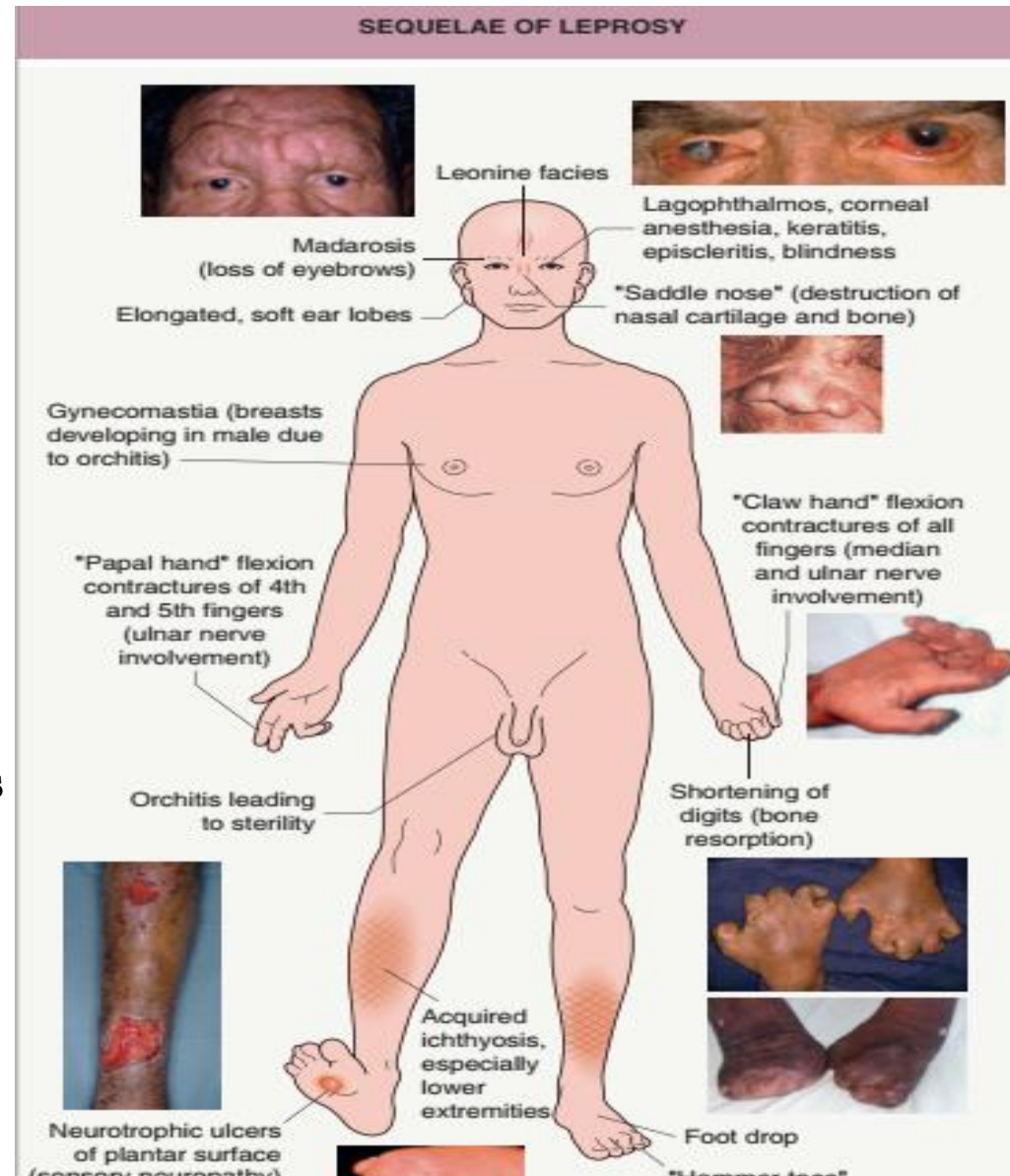
Claw hand → median & ulnar → all fingers

Hammer toes:- foot drop

Glove & stocking anaesthesia

Acquired ichthyosis

Testicular atrophy



H.P.

stains: fite stain → red

sudan 3 & 4

methamine silver stain → organism fragments

Tubercloid

- Well defined tubercloid granuloma.
- linear pattern with few liprae bacilli

lepomatous

- Diffuse tubercoid granuloma
- Granz zone
- mini bacilli → foamy appearance of Virchow cells.

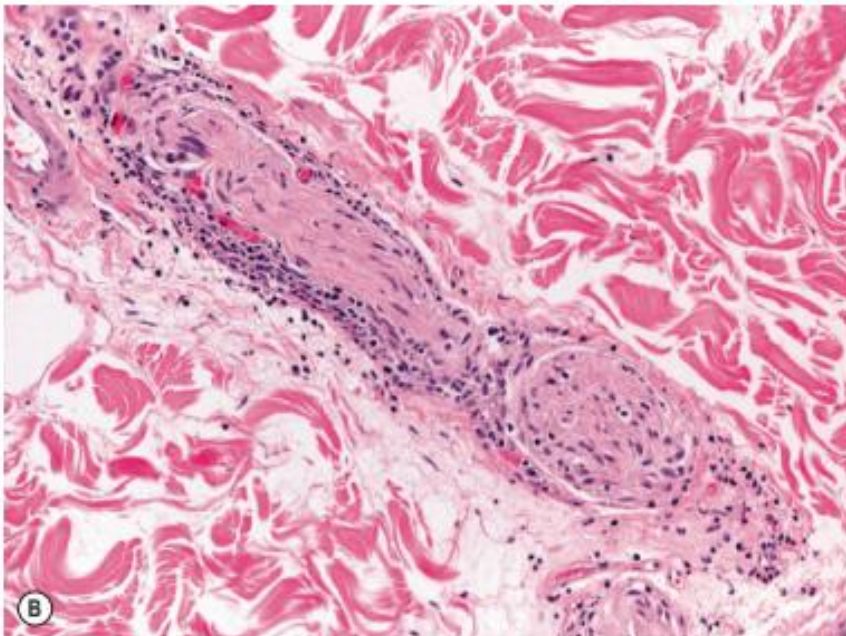
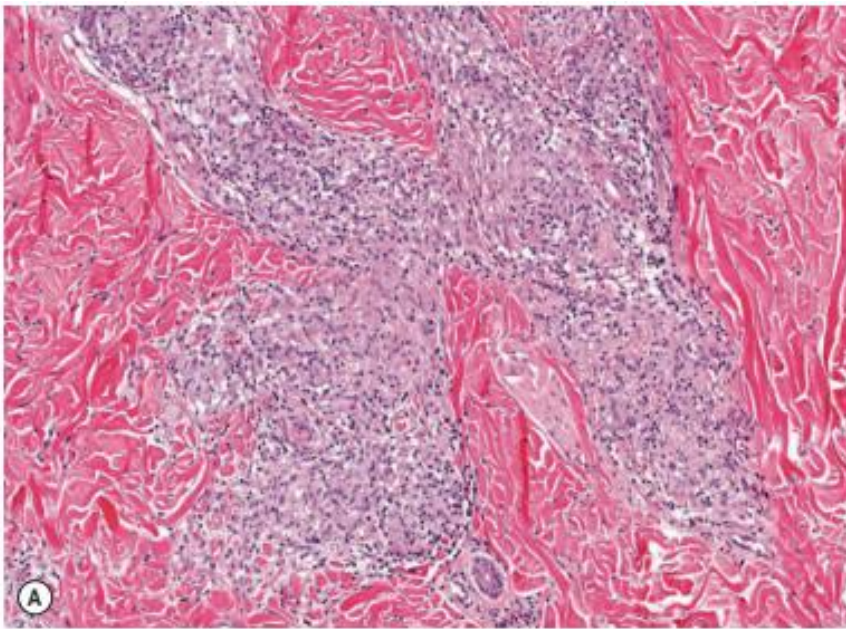


Fig. 75.10 Tuberculoid leprosy. **A** Granulomatous nodular infiltrate within the dermis. **B** Nerve involvement characterized by a sparse perineural infiltrate with lymphocytes, histiocytes and plasma cells. *Courtesy, Lorenzo Cerroni, MD.*

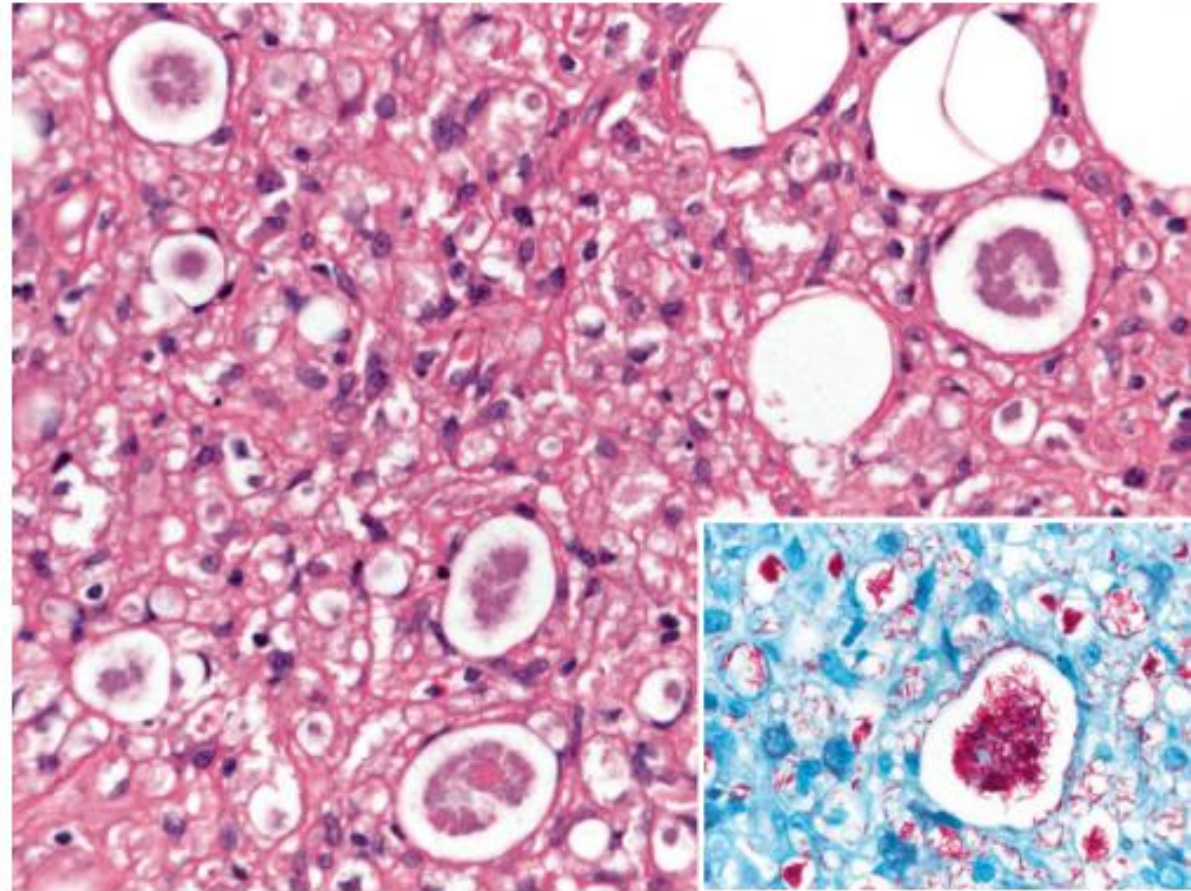


Fig. 75.9 Lepromatous leprosy. Diffuse dermal infiltrate of foamy histiocytes; note the globi which represent clumping of mycobacteria. Fite stain highlighting the numerous bacilli as well as globi (inset). *Courtesy, Lorenzo Cerroni,*

- **Borderline BB,BT and BL**

Depend on position of pt. on spectrum →
asymmetric macular plaque → punched out, swiss
cheese & island of normal skin

Nerve affection precedes skin lesion



Fig. 75.6 Borderline leprosy. Erythematous plaques with central clearing leading to annular (A) and arcuate (B) lesions. Note the ill-defined outer borders and well-demarcated inner borders.

Special types of leprosy

- Pure neurotic:- asymmetrical with peripheral nerve affection with no visible skin lesion → onion skin appearance.
- Lucio leprosy:- diffuse thickening of skin.
- Histoid LL:- dermatofibroma like
- Indeterminant :- early skin lesion

Investigations

A] organism determination:-

Biopsy

Slit skin smear

→ 6 routine sites ear lobes, elbows and knee → skin fold gripped firmly → incision
→ fluid & dermal tissue → air dried never heat fixed.

Bacterial index

Morphological index

PCR



- Determine immunity:-

Lepromine test

→ ID injection of 0.1 ml heat killed *M. lepra* → fernandx reaction → after 48 hours in LL & TL

mitcudal reaction → after 4 weeks with TL only

Anti PGL 1 [phenolic glycolipid 1]

False positive syphilis test

VRDL, FTA & ABS

ttt

Paucibacillary	multibacillary
Rifampicin 600 mg monthly Dapson 100 mg daily for six month Follow up for 2 ys.	Rifampicin 600 mg monthly Dapson 100 mg daily Clofazimine 300 mg monthly 50 mg daily For 12 months Follow up for five ys

Peuci single lesion → rom regimen → rifampicin, ofloxacillin & minocyclin

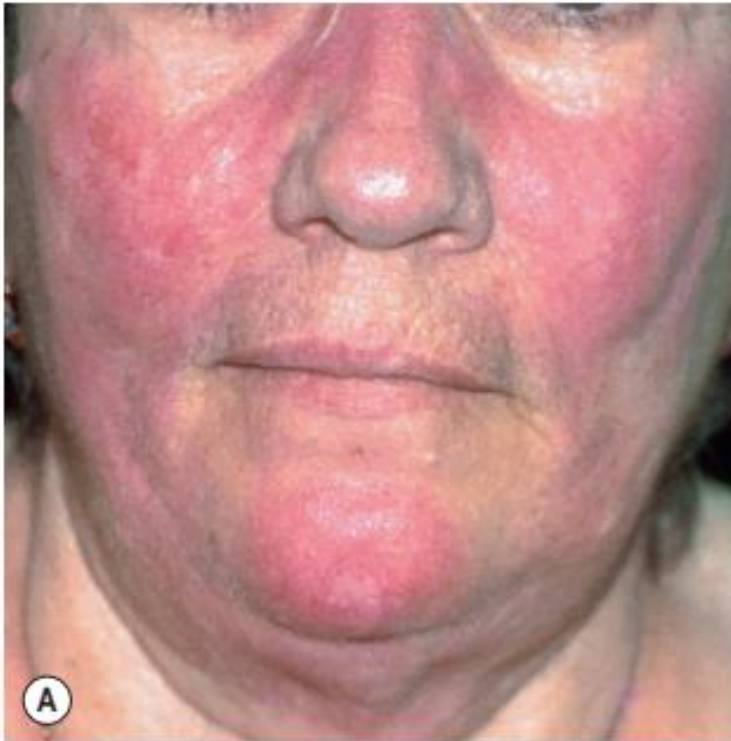
MULTIDRUG THERAPY/WHO SCHEME FOR THE TREATMENT OF LEPROSY						
	Rifampin	Clofazimine	Dapsone	Ofloxacin	Minocycline	Therapy duration
MB (>5 lesions*)	600 mg once monthly	300 mg once monthly and 50 mg daily	100 mg daily	–	–	12 blister packs over 12 to 18 months
PB (2–5 lesions*)	600 mg once monthly	–	100 mg daily	–	–	6 blister packs over 6 to 9 months
PB (single lesion*)	600 mg × 1	–	–	400 mg × 1	100 mg × 1	Single dose
Dose adjustments for children						
10 to 14 years of age, MB [†]	450 mg once monthly	150 mg once monthly and 50 mg every other day	50 mg daily	–	–	12 blister packs over 12 to 18 months
<10 years of age, MB [†]	300 mg once monthly	100 mg once monthly and 50 mg twice weekly	25 mg daily	–	–	12 blister packs over 12 to 18 months
5 to 14 years of age, single lesion PB	300 mg × 1	–	–	200 mg × 1	50 mg × 1	Single dose

Lepra reaction

Reveral neuritis	Erythema nodosum deprosum	Lucio reaction
• Borderline (BB, BT & BL) • Increased CMI → increase TH 1 • Type IV hypersensitivity • Acute neuritis • Inflamed skin lesion +/- new skin lesion • No constitutional • No ulcer • Edema • Ttt → prednisolone 40 mg	LL & BL ++ humoral immunity Type III hypersensitivity Immune complex → vasculitis Painful red nodule in face Limb ulceration Constitutional Uveitis , neuritis, arthritis & myositis Ttt → thalidomide 400 mg	Lucio [diffuse LL] Necrotizing vasculitis Erythematous patch → deep ulcer → fatal

TWO MAJOR TYPES OF LEPROSY REACTIONAL STATES

	Type 1 – reversal reaction with “upgrading”	Type 2 – vasculitis (most commonly erythema nodosum leprosum)
Immune mechanism	Enhancement of cell-mediated immunity with a Th1 cytokine pattern	Excessive humoral immunity with a Th2 cytokine pattern and formation of immune complexes; may be accompanied by increased cell-mediated immunity
Pathogenic process	Delayed-type hypersensitivity reaction	Cutaneous and systemic small vessel vasculitis
Leprosy category	Borderline (BT, BB, BL) or tuberculoid with immunologic recovery during or after treatment	Lepromatous and BL > BB; especially patients with a high bacterial index who are undergoing treatment
Clinical characteristics	• Increased inflammation in established skin lesions • Emergence of “new” skin lesions • Acute nerve pain or tenderness (neuritis) and loss of function • Recent (<6 month history) or progressive neurologic impairment in the absence of painful nerves	• Nodular skin lesions • Fever, myalgias, malaise • Severe joint swelling and pain • Iridocyclitis • Lymphadenitis • Hepatosplenomegaly • Orchitis • Glomerulonephritis
Treatment	Prednisone	Thalidomide



SOE Dermatology / EBDV by dr. Eman Gomaa -
Connect Academy

Fig. 75.8 Leprosy reactions. **A** Type 1 "upgrading" reaction with lupus erythematosus-like facial involvement in a "butterfly" distribution with marked inflammation. **B** Erythema nodosum leprosum, a type 2 reaction, presenting with erythematous nodules on the legs. **C** Type 2 reaction presenting with red facial papulonodules due to immune complex-mediated small vessel vasculitis in patients with lepromatous leprosy. *C, Courtesy, Louis A Fragola, Jr, MD.*



eFig. 75.3 Type 1 leprosy reaction. "Upgrading" reaction in a patient with borderline lepromatous disease characterized by marked inflammation of a facial

Clinical criteria of leprosy

1] clinical

Typical skin lesion

Anaesthesia

Thickened nerve

2] SSS → AFB

3] H.P.

DD

- Hypo pigmented macule?
- Erythematous plaque?
- Lepra reaction?

CLINICAL DIFFERENTIAL DIAGNOSIS OF LEPROSY

Hypopigmented lesions	• Mycosis fungoides, including the hypopigmented variant • Sarcoidosis, but the hypopigmented variant is usually more papulonodular • Postinflammatory hypopigmentation
Circinate (annular) plaques	• Mycosis fungoides and other forms of cutaneous lymphoma • Sarcoidosis • Psoriasis • Interstitial granulomatous dermatitis • Granuloma annulare • Tinea corporis
Infiltrated plaques/nodules	• Lymphoma • Sarcoidosis • Other infections (e.g. dimorphic fungal infection, tuberculosis, leishmaniasis, syphilis) • For DDx of leonine facies, see Table 46.5

Neurologic findings	• Peripheral neuropathy due to other disorders (e.g. diabetes mellitus, vasculitis, nutritional disorders, syringomyelia)
Deforming acral features	• Systemic sclerosis • Tabes dorsalis • Dupuytren contracture
Type 1 reaction	• Acute cutaneous lupus erythematosus • Cellulitis and pseudocellulitis • Drug reaction • Misdiagnosis as worsening of disease when subclinical lesions become apparent
Type 2 reaction	• Sweet syndrome • Medium-vessel vasculitis • Panniculitides, e.g. erythema nodosum • Infections, including other mycobacterial, bacterial, dimorphic fungal, and other opportunistic fungal

Tuberculosis

- *M. tuberculosis*

Acid & alcohol fast

3 species → *M. tuberculosis*, *M. bovis* & BCG attenuate weak strain

Route:- inhalation, ingestion & inoculation



epdemiology

- 1/3 of the world
- ++ by HIV, resistance, TNF alpha -- & immunosuppression
- Africa, India & China

Cutaneous T.B.

True TB

Exogenous

- 1] TB chancre [non immune]
- 2] wart TB [immune] +/-
- 7] lupus vulgaris

Endogenous

- 3] Scrofuloderma [contagious]
- 4] orofacial [auto inuclation]
- 5] military T.B
- 6] T.B gamma

Tubercloid

- 1] lichen scrofulorum
- 2] papulonecrotic
- 3] erythema induratum of Bazin



TB CHANCERE



Fig. 75.13 Tuberculosis verrucosa cutis. A wart-like papule at the site of exogenous inoculation in a patient with immunity against *M.*



Figure 5. Tuberculosis verruca cutis showing verrucous



Fig. 75.14
Scrofuloderma. Plaques and nodules with central ulceration as well as resultant scarring with retraction.



Fig. 75.15 Orificial tuberculosis. A non-healing ulcer of the nasal mucosa.
Courtesy, Louis A Fragola, Jr, MD.

SCROFULODERMA





MILIARY TB



TB GUMMA

LUPUS VULGARIS



Fig. 75.16 Lupus vulgaris. A Annular granulomatous plaque with central scarring. B Coalescing pink-brown papules. C Red-brown plaque on the neck. C, Courtesy, Eugene Mirre, MD.



Etiology of cutaneous TB

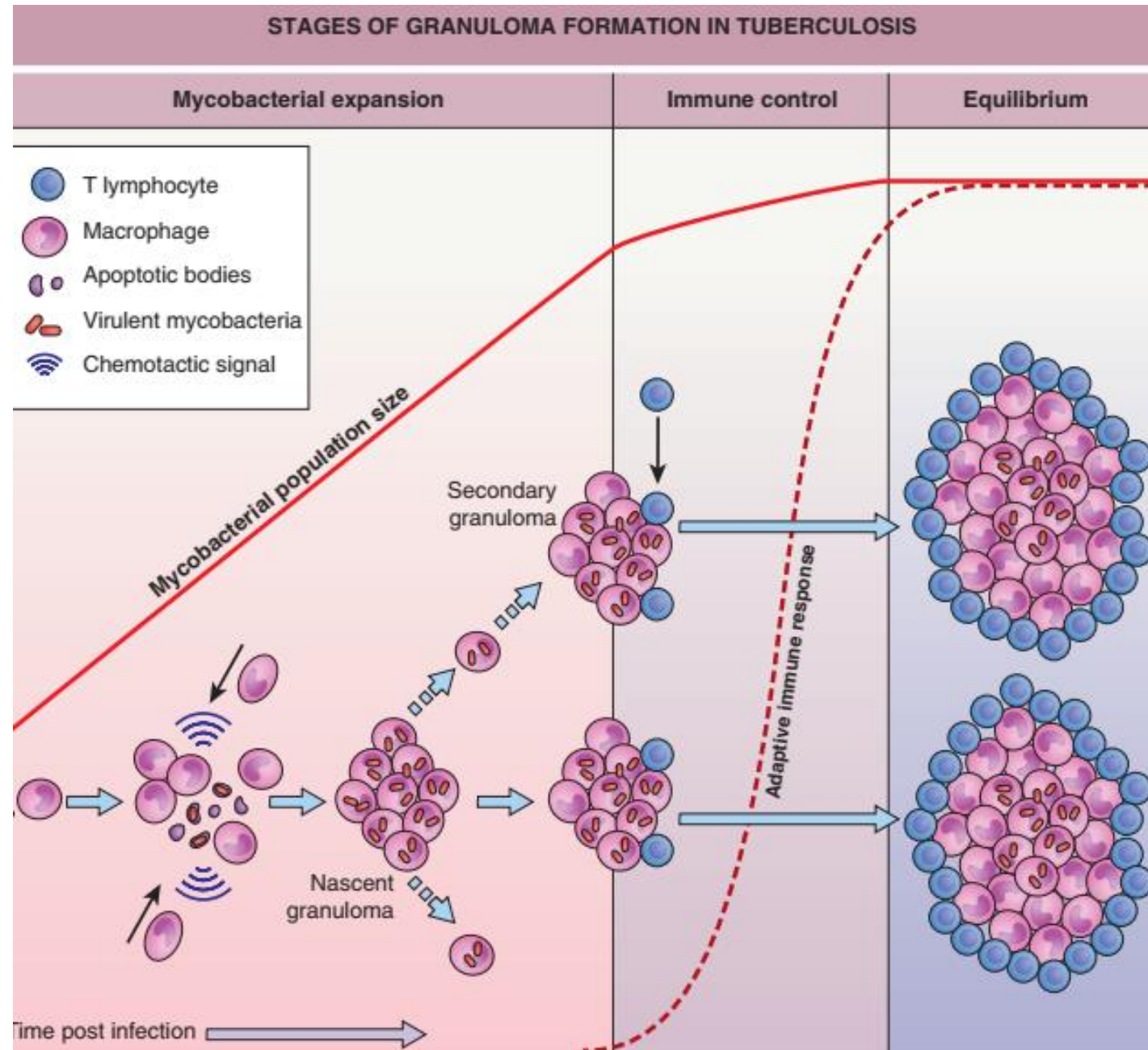
- M tuberculosis → by direct inoculation, hematogenous & underlying structure
- Epidemiology:- rare → 1-2% of infected patients

• Pathogenesis:-

1] mycobacterial expansion →
multiplication →
phagocytosed → resist
intracellular
digestion. → nascent granuloma

2] immune control →
PRR, TLR → CD4

3] stage of equilibrium

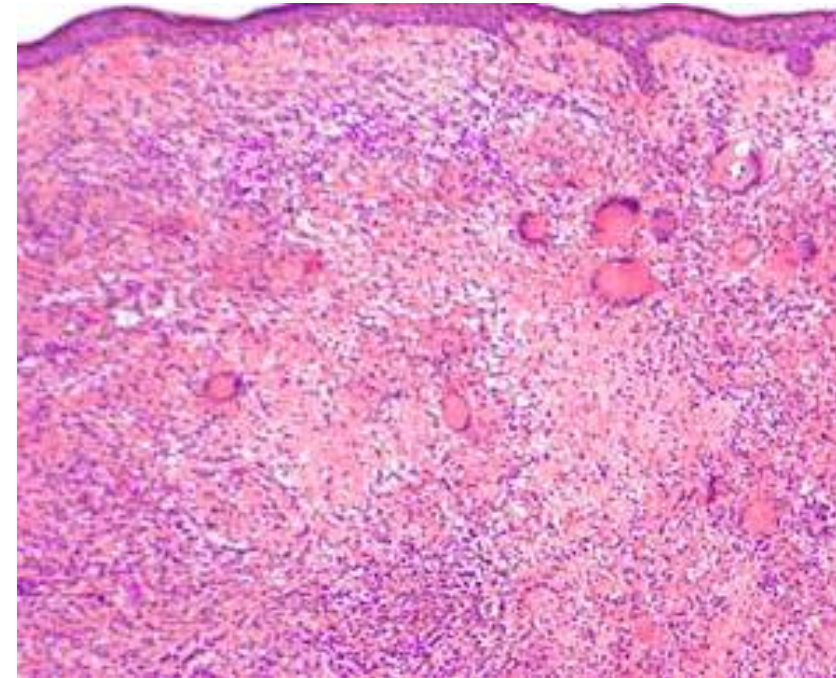
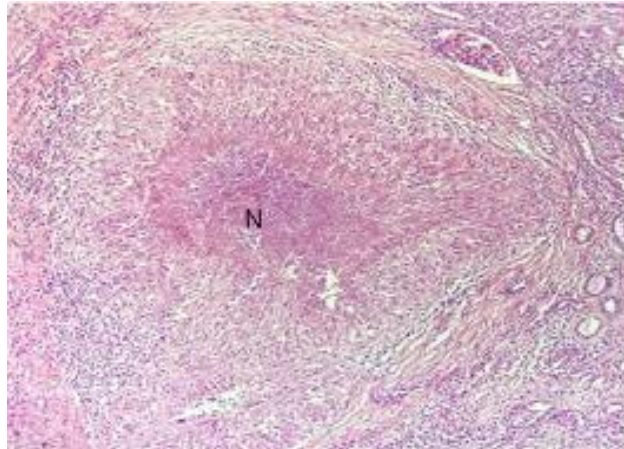


- Investigations :-

H.P. → early non- specific & late tubercloid granuloma → acid fast bacilli with ziehl Neelsen stain

Cultur → lewenstien Jensen media

PCR



Tuberclin test → ID injection of PPD → type IV hypersensitivity.

Methods →

the mantoux test

& heaf test



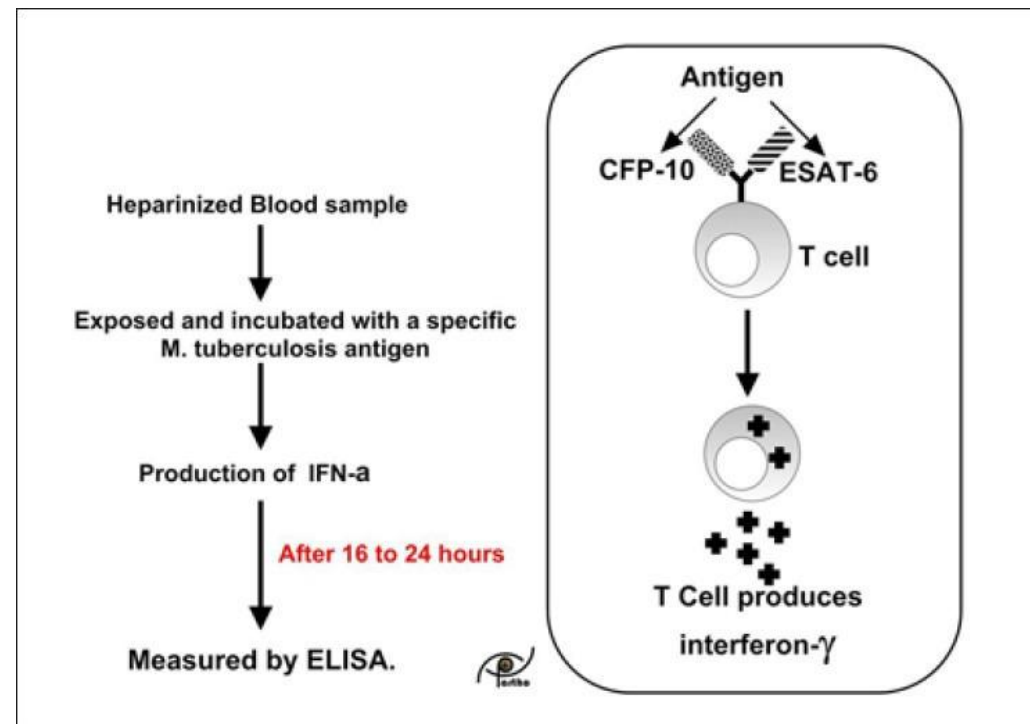
- Interferon gamma release assay

T cell blood test [more specific]

ESAT 6

Quantiferon TB

T spot TB



ADVANTAGES AND DISADVANTAGES OF IFN- γ RELEASE ASSAYS AND TUBERCULIN SKIN TESTING

	Tuberculin skin test	IFN- γ release assays
Advantages	• Does not require a laboratory • Relatively low cost	• Requires a single patient visit • Results can be available within 24 hours • Prior BCG vaccination does not cause a false-positive result • Does not boost responses measured by subsequent tests
Disadvantages	• Requires two patient visits • Results are not available for 48 hours • Prior BCG vaccination can cause a false-positive result • May boost responses in subsequent tests • Infections with non-tuberculous mycobacteria may lead to a false-positive result	• Requires laboratory processing within 16 hours (for QuantiFERON® TB Gold [In-Tube]) or 8–30 hours (for T-SPOT®. TB; the latter if using T-cell Xtend®) • Relatively high cost • Infections with some non-tuberculous mycobacteria (e.g. <i>M. kansasii</i>, <i>M. szulgai</i>, <i>M. marinum</i>) may lead to a false-positive result

- Prevention

BCG vaccine ;

Compulsory vaccine in →

1st 15 days of life

Dose to prevent skin TB



- Ttt

A] general measure :-

Notification

Bacteriological exam

HIV testing [TB is AIDS defining illness]

B] Standard drug regimen [6 months]

Isoniazid 300 mg

Rifampicin 600 mg

For 6 months

Pyrazinamide & ethambutol 15 mg/kg for 2 months

C] Modification for dermatology :-

Excision & plastic surgery

D] newer drugs

Bedacholine [sirturo]

MVA vaccine

CLASSES, ACTIVITIES AND CROSS-RESISTANCES OF THE ANTITUBERCULOUS AGENTS			
Agent	Class of compound	Spectrum of activity	Cross-resistance to other antituberculous agents
First-line agents			
Rifampin	Antibiotic	Broad	Other rifamycins
Isoniazid	Synthetic	Tubercle bacilli	None
Pyrazinamide	Synthetic	Tubercle bacilli	None
Ethambutol	Synthetic	Tubercle bacilli	None
Rifapentine	Antibiotic	Broad	Other rifamycins
Rifabutin	Antibiotic	Broad	Other rifamycins
Second-line agents			
Kanamycin, amikacin, streptomycin	Aminoglycoside antibiotics	Broad	Other aminoglycosides, viomycin, capreomycin
Capreomycin, viomycin	Tuberactinomycin antibiotics	Tubercle bacilli	Aminoglycosides
Levofloxacin, ofloxacin, moxifloxacin	Quinolone antibiotics	Broad	None
Ethionamide, prothionamide	Thioamides	Tubercle bacilli	Thiacetazone
Cycloserine, terizidone	Synthetic	Broad	None
<i>para</i> -Aminosalicylic acid	Synthetic	Tubercle bacilli	None
Third-line agents			
Clofazimine	Riminophenazine antileprotic	Mycobacteria	None
Amoxicillin/clavulonate	β -lactam antibiotic/ β -lactamase inhibitor	Broad	None
Linezolid	Oxazolidinone antibiotic	Broad	None
Imipenem	β -lactam antibiotic	Broad	None
Thiacetazone	Synthetic	Tubercle bacilli	Ethionamide, prothionamide
Clarithromycin	Macrolide antibiotic	Broad	None
Bedaquiline	Diarylquinoline antibiotic	Broad	None
Delamanid	Dihydro-nitro-imidazooxazole	Mycobacteria	None

MULTIDRUG-RESISTANT AND EXTENSIVELY DRUG-RESISTANT TUBERCULOSIS

Multidrug-resistant tuberculosis (MDR-TB)

- Resistant to isoniazid and rifampin, the two most effective first-line drugs
- Estimated to account for ~3% of all new TB cases worldwide in 2015, but 20% of individuals previously treated for TB

Extensively drug-resistant tuberculosis (XDR-TB)

- Resistant to isoniazid and rifampin
- Resistant to quinolones, the most effective second-line drugs
- Resistant to at least one of three injectable second-line drugs: amikacin, kanamycin, or capreomycin
- Estimated to account for 10% of cases of MDR-TB

	lupus vulgaris [unhealthy scar]
Def	Paucibacillary form of cutaneous TB → in previous desensitized Pt → with strong +ve delayed hypersensitivity
Epidem.	5-10 % of cutaneous TB Female > male
Way of infection	Endogenous, exogenous, haematological, direct extension from underlying structure, on BCG vaccine, on scar of wart TB & scrofuloderma
C/pic	Red brown plaque → with vegetated border → central scarring Diascopy → apple jelly nodule In nose, cheeks & ear lobules 5 variants Mucosal [nasal, buccal & conjunctiva] affection

H.P	Well developed tubercloid granuloma Scanty or absent caseation
Complications	Scarring + tissue destruction +contracture + malignancy [SCC]
Tuberclin test	+ve
DD	Sarcoidosis, DLE, LL & leishmania
prognosis	Progressive if untreated

POC	TB Chancre	Wart TB
Def	Exogenous TB in non immune subject	In immune subjects
causative	M. Bovis in HIV	
c/pic	Painless firm papule & nodule → chronic painless ulcer in face & limbs	Asymptomatic painless warty like papule → verrucous plaque → in hand, buttocks, knee, elbow & ankle
H.P:-	Tubercloid granuloma [multibacillary]	Pseudoepitheliomatous hyperplasia & paucibacillary
TST	Early – late + [non immune early]	+ ve from the start
DD	Chronic ulcer	wart

	Scrofuloderma Commonest form in children	Orofacial TB
Def:-	Endogenous contagious Skin overlying TB focus → bone & LN.	Endogenous auto inuclation → adjacent orifices → mouth, tongue, anus, vulva & penis
c/pic	Painless SC nodules → cold abscess → ulcer → sinus track formation	Painful papule → ulcer
H.P.	Diffuse granuloma → caseation [multibacillary]	Diffuse granuloma → caseation [multibacillary]
TST	- ve	- ve
DD	Hydradenitits suppruativa	Oral ulcer

	Miliary TB	TB gamma
Def	Endogenous → hematogenous VVVV low immunity	Endogenous → hematogenous
c/pic	Pin head bluish-red papules → vesicles → crustations → white scars all-over the body	Single or multiple firm SC nodule → painless ulcer in extremities > trunk
H.P.	Nonspecific inflammatory infiltrate (multibacillary)	Massive necrosis (multibacillary)
TST	- ve	- ve
DD	PLEVA & varicella	Paniculitis & syphilitic gamma

CLINICAL PRESENTATIONS OF CUTANEOUS TUBERCULOSIS

	Tuberculous chancre	Tuberculosis verrucosa cutis	Scrofuloderma	Orificial tuberculosis	Lupus vulgaris	Acute miliary tuberculosis	Tuberculous gumma
Synonyms	Primary inoculation tuberculosis, cutaneous primary complex	Warty tuberculosis, prosector's wart, lupus verrucosus	Tuberculosis cutis colliquativa	Acute tuberculous ulcer, tuberculosis cutis orificialis	Tuberculosis luposa	Tuberculosis cutis disseminata, tuberculosis cutis acuta generalisata	Metastatic tuberculous ulcer, metastatic tuberculous abscess
Incidence	Rare except in Asia; 1–2% of cutaneous tuberculosis	Most common form of cutaneous tuberculosis (e.g. ~40% of cases in Hong Kong)	Common in immigrants from developing countries	Rare	Low: 10–15% of cutaneous tuberculosis; women affected 2–3 times more often than men	Rare; infants and young children, seldom adolescents or adults	Children with low socioeconomic status or immunocompromised hosts
Transmission	Inoculation into skin and/or mucosa of previously uninfected person, i.e. with no specific immunity	Inoculation into skin and/or mucosa of previously infected person with moderate or high immunity	Contiguous involvement of skin overlying a focus of tuberculosis, usually in lymph node or bone. Also reported after BCG vaccination	Autoinoculation of mucosa or skin adjacent to a natural orifice draining an active internal tuberculous infection	Direct extension; hematogenous or lymphatic spread from a tuberculous focus; reinfection; BCG vaccination	Hematogenous dissemination from primary lung focus in patient with low immunity; may follow viral infection	Acute hematogenous dissemination from a primary focus during periods of bacillema and lowered resistance

Tuberculin skin test	Early: negative Later: positive	Positive	Usually positive	Variable, often negative	Usually positive	Usually negative	Usually negative
Histopathology	- Acute: nonspecific neutrophilic inflammation with necrosis; bacilli - Later: granulomatous inflammation with central caseation, epithelioid and Langhans giant cells; disappearance of bacilli	Acute inflammation in the epidermis, pseudoepitheliomatous hyperplasia, microabscesses in upper dermis, sparse granulomatous foci; occasionally bacilli present	Tuberculous granulation tissue and caseation necrosis in the deeper dermis; bacilli can be isolated from pus	Nonspecific inflammatory infiltrate and necrosis, some tubercles with caseation deep in dermis; bacilli easily found	Well-developed tubercles with scant caseation, nonspecific inflammatory infiltrate; no visible bacilli	Necrosis, nonspecific inflammatory infiltrate surrounded by macrophages, sometimes developing into small abscesses; abundant bacilli	Massive necrosis and abscess formation; large number of bacilli
Diagnosis	Histology and culture PCR IGRA*	Culture PCR IGRA	Culture PCR IGRA	Histology and culture PCR Presence of another focus of TB IGRA*	Histology and culture PCR IGRA	Smear, histology and culture; PCR Presence of another focus of TB IGRA*	Culture and histology PCR IGRA*
Differential diagnosis	Other pyogenic ulcers and chronic chancroid lesions, including the following: sporotrichosis, dimorphic or opportunistic fungal infections, <i>M. marinum</i> infection,	Warts, verrucous fungal infections (e.g. chromomycosis), tertiary syphilis, hypertrophic lichen planus	Dimorphic or opportunistic fungal infections, non-tuberculous mycobacterial infections, syphilitic gumma, actinomycosis, chronic bacterial osteomyelitis, adenitis with drainage,	Painful ulcerative diseases that affect orifices and periorificial skin: HSV, recurrent aphthous stomatitis, pemphigus, histoplasmosis	Sarcoidosis, discoid lupus erythematosus, dimorphic fungal infections, leishmaniasis, tertiary syphilis, squamous cell carcinoma	Varicella, enteroviral exanthems, rickettsialpox, PLEVA	Panniculitis, dimorphic or opportunistic fungal infection, syphilitic gumma, hidradenitis suppurativa

Tuberclids

- Cutaneous hypersensitivity reaction to hematogenous M. TB antigen → strong cell mediated immunity
- Diagnostic criteria
 - Bilateral, symmetrical
 - ve culture
 - H.P. → tubercloid granuloma without bacilli
 - +ve TST
 - Improve by anti-TB drugs
 - Past TB

Lichen scrofulusorum	papulonecrotic	Erythema induratum of Bazin
Asymptomatic lichenoid papules Trunk → annular configuration → children	Papulo-pustules → central necrosis → extensors	Tender painful dusky nodules → ulcerate → in calves



Fig. 75.17 Papulonecrotic tuberculid. Erythematous papules and papulopustules on the heel.



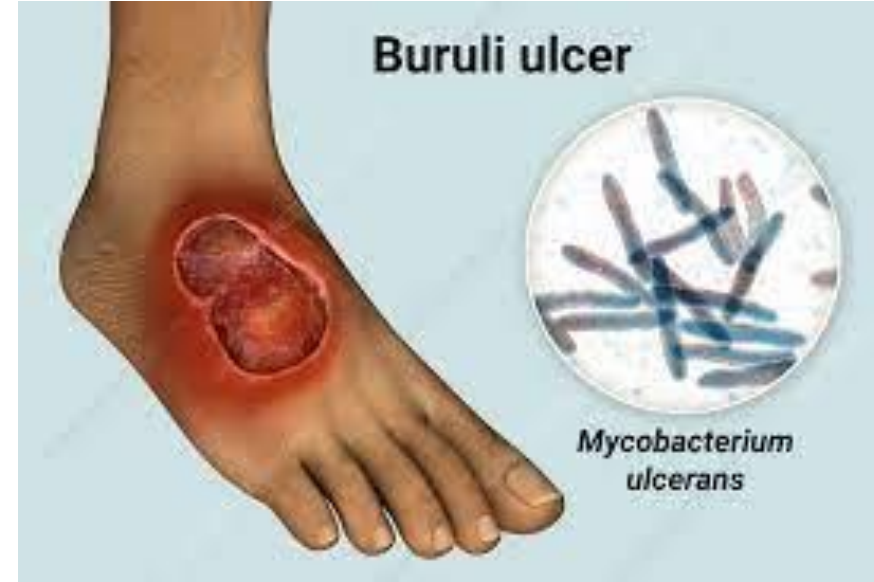
non-tuberculous mycobacterium

Buruli ulcer	Swimming pool granuloma
M. Ulcerans Fresh water [snail] LL Firm nodule → ulcer → no lymphatic Treated by local heating & rifampicin	M. Marinum Salt water UL Nodule → ulcer → lymphatic [Sporotrichoid] Clarithromycin & doxycycline





Fig. 75.21
Mycobacterium marinum infections. A range of presentations, including an erythematous plaque with scale-crust at the inoculation site on the lateral hand (A), a sporotrichoid pattern with the inoculation site on the distal third finger (B), and disseminated necrotic lesions on the face of an immunocompromised patient



Fish tank granuloma



TREATMENT OF NON-TUBERCULOUS ("ATYPICAL") MYCOBACTERIAL INFECTIONS

Agent	Local	Systemic medications: alone or combined
<i>M. ulcerans</i>	For ulcers, surgical excision; local heat (40°C); hyperbaric oxygen	Rifampin plus streptomycin, clarithromycin, and/or a quinolone (e.g. moxifloxacin, ciprofloxacin) for at least 4–8 weeks*
<i>M. marinum</i>		Clarithromycin, minocycline > doxycycline [†] , rifampin, ethambutol, trimethoprim–sulfamethoxazole; two agents for serious infections (e.g. clarithromycin + rifampin or ethambutol)
<i>M. kansasii</i>		Isoniazid (or clarithromycin) + rifampin + ethambutol ± pyridoxine; other options (depending on susceptibilities) include azithromycin, moxifloxacin, streptomycin and sulfamethoxazole
<i>M. fortuitum</i>	Surgical excision	Ciprofloxacin, ofloxacin, clarithromycin [‡] , trimethoprim–sulfamethoxazole, minocycline, doxycycline; at least two agents for serious infections
<i>M. chelonae</i>	Surgical excision	Clarithromycin; for severe infections, combine clarithromycin with linezolid, imipenem, tobramycin, amikacin, doxycycline and/or ciprofloxacin (often resistant to latter two agents)
<i>M. abscessus</i>	Surgical excision	Clarithromycin or azithromycin + (for serious infections) amikacin, cefoxitin or imipenem
<i>M. avium</i> complex		Clarithromycin or azithromycin + ethambutol ± rifampin or rifabutin
<i>M. haemophilum</i>	Surgical excision	Clarithromycin + rifampin ± ciprofloxacin or amikacin
<i>M. scrofulaceum</i>	Surgical excision	Isoniazid + rifampin, clarithromycin



THANK YOU